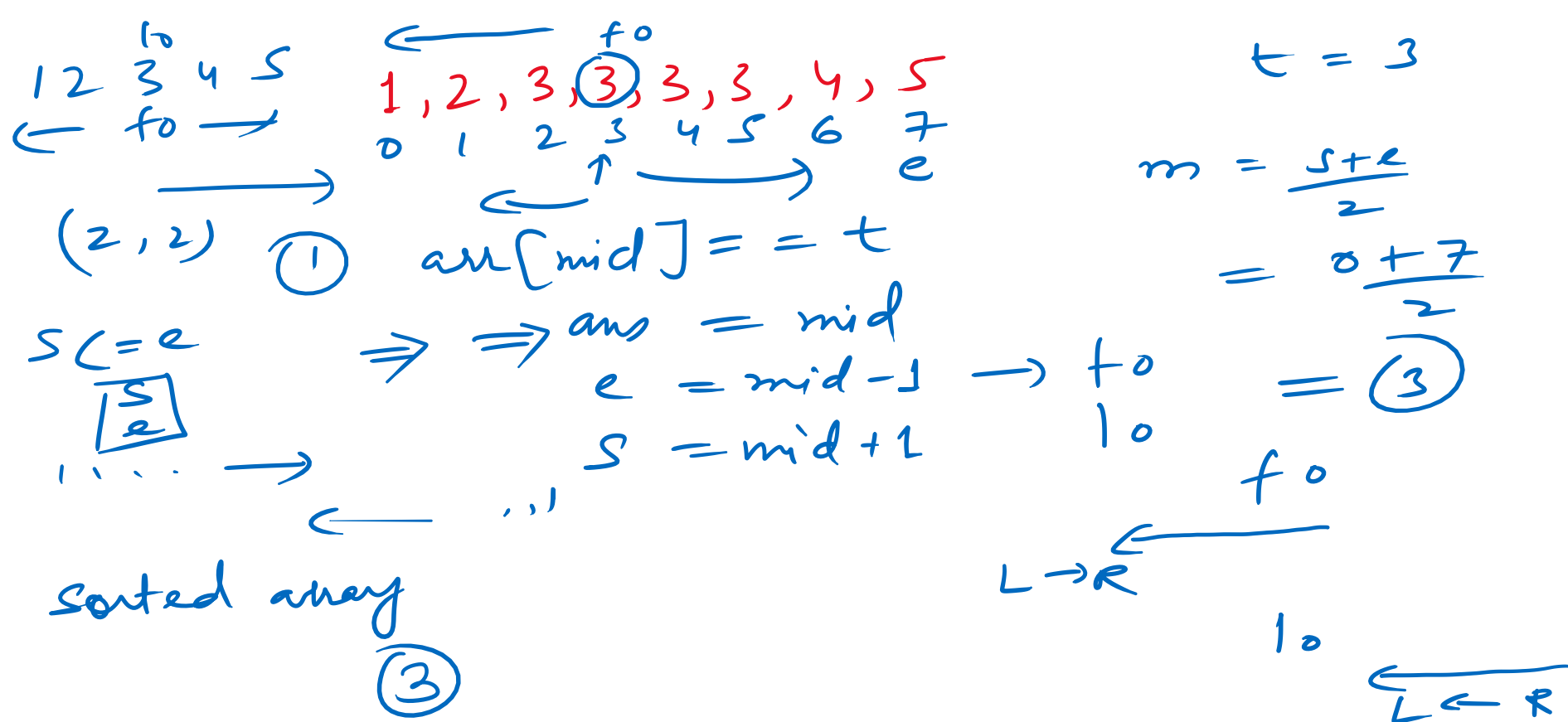




Applications of Binary Search:

* Finding the square root of a given number.

$ans = mid = 3$ $n = \sqrt{36} \rightarrow 6$ [Search Space]

$0 \quad \frac{+}{18} \quad 36$
 $s \quad \quad e$
 $e = m - 1 = 18 \rightarrow 17$
 $m = \frac{s+e}{2} = \frac{18}{2} = 9$
 $m \times m = 18 \times 18 = 324$
 $0 \quad \frac{+}{8} \quad 17$
 $\leftarrow$
 $0 \quad \quad 7$
 $S = m + 1$
 $4 \quad \frac{+}{5+1} \quad 7$
 $\frac{11}{2} = 5$
 $5 \times 5 = 25 < 36$

$ans = 5$

$S = m + 1$

$6 \quad \quad 7$

$6 + 7 = \frac{13}{2} = 6$

$6 \times 6 = 36$

$m \times m < n$
 $ans = mid$
 $S = mid + 1$

$m \times m > n$
 $e = mid - 1$

Search Space
 $0 \quad \quad 1024$

512

256

$6 \cdot 012$

return ans; bspre (37, 3)

* Largest Element Less than or Equal to a Target

$2 \ 4 \ 10 \ 19 \ 23 \ 45 \ 87$
 $\leftarrow \rightarrow$

$t = 20$

①9

* Smallest Element Greater than or Equal to a Target.

$2 \ 4 \ 10 \ 19 \ 23 \ 45 \ 87$
 $s \quad \quad e$

$t = 20$

②3